

AI & ML Task 2

Feature Engineering, Model Optimization and Performance Comparison

Name: AKSHAT UPADHAYA

1. Introduction

Machine Learning models are widely used to make predictions based on historical data. The performance of these models depends not only on the algorithm but also on the quality of data preprocessing and feature engineering techniques.

The objective of this project is to compare the performance of different regression models on the California Housing Dataset. The project includes data preprocessing, feature scaling, model training, evaluation, and comparison to identify the best-performing model.

2. Methodology

2.1 Dataset Description

The California Housing Dataset provided by Scikit-Learn was used for this project. The dataset contains housing-related information collected from various districts of California.

Features Used

- MedInc (Median Income)
- HouseAge (Median House Age)
- AveRooms (Average Rooms)
- AveBedrms (Average Bedrooms)
- Population
- AveOccup (Average Occupancy)
- Latitude
- Longitude

Target Variable

- HousePrice (Median House Value)

The dataset contains 20,640 records and does not contain any missing values.

2.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the dataset and identify relationships among features.

The following visualizations were created:

- House Price Distribution
- Correlation Heatmap
- Median Income vs House Price Scatter Plot
- Boxplots Before and After Feature Scaling

The correlation analysis showed that Median Income (MedInc) has the strongest positive relationship with house prices.

2.3 Feature Engineering

Before training the models, feature scaling was applied using the StandardScaler technique. Feature scaling transforms all features to a common scale with a mean of zero and a standard deviation of one. This helps machine learning algorithms perform more effectively and ensures that no feature dominates because of its larger numerical values.

2.4 Model Training

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

Three machine learning regression models were trained and evaluated:

1. Linear Regression
2. Ridge Regression
3. Decision Tree Regressor

These models were selected because they represent different approaches to regression and provide a good basis for performance comparison.

3. Results and Discussion

The performance of each model was evaluated using the following metrics:

Evaluation Metrics

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Model Comparison

Model	MAE	RMSE	R^2 Score
Linear Regression	0.7456	0.5332	0.5758
Ridge Regression	0.7456	0.5332	0.5758
Decision Tree Regressor	0.7242	0.5223	0.5997

Observations

- Lower RMSE values indicate better prediction accuracy.
- Higher R^2 scores indicate better model performance.
- Feature scaling improved model stability and ensured fair comparison among algorithms.
- The model with the highest R^2 score was selected as the best-performing model.

The Actual vs Predicted graph showed that many predicted values were close to the actual values, indicating that the model captured the general trend of the housing data.

The residual plot showed that prediction errors were distributed around zero, suggesting reasonable model performance.

4. Conclusion

In this project, multiple regression models were trained and compared using the California Housing Dataset.

The dataset was preprocessed and standardized using StandardScaler. Exploratory Data Analysis helped identify important relationships among variables, with Median Income emerging as the most influential feature affecting house prices.

Three regression models were evaluated using MAE, RMSE, and R² Score. Based on these metrics, the best-performing model was identified and selected.

This project helped in understanding data preprocessing, feature scaling, model training, model evaluation, and comparative analysis of machine learning algorithms. The knowledge gained through this task provides a strong foundation for developing more advanced machine learning models in future projects.

Future Scope

- Hyperparameter Tuning
- Random Forest Regression
- Gradient Boosting Regression
- XGBoost Regression
- Advanced Feature Engineering

These techniques may further improve prediction accuracy and overall model performance.

